

LIGHTWEIGHT SAFE HELICOPTER (LSH)

DOCUMENT 00 — COMPREHENSIVE ENGINEERING BASELINE — HYBRID HELIUM

Generation 1 V4.0 Variant — Hybrid Helium

1. Status and Engineering Basis

Status: [PROPOSED] Preliminary controlled conceptual engineering baseline. This document is a variant study derived from the LSH V4.0 Gen1 baseline. Numerical values are targets, assumptions or preliminary calculations and are not demonstrated performance, certification results or flight-proven values.

The common LSH survival philosophy is retained: automatic detection, independent backup initiation, independent manual crew authority, deterministic Safety Kernel authorization, selective physical separation, occupant survival, recovery and post-event survival. Variant-specific hazards and failure modes must be analyzed separately.

The current LSH V4.0 baseline is a 20,000 kg-class single-main-rotor aircraft with a 21.0 m main rotor, 3.5 m tail rotor, 1,000 km range target, occupied survival capsule and eight fixed short supports. This variant changes the lift/propulsion architecture and therefore requires independent mass, CG, dynamics, structures, performance and certification closure.

2. Top-Level Technical Baseline

Parameter	Variant baseline	Status
Design / operating mass target	20,000 kg	TARGET
Architecture	Rotorcraft + integrated non-rigid/semi-rigid helium buoyancy envelope	CONCEPT
Helium gross volume	≈ 7,540 m ³ for ~40% gross static weight support at sea-level reference density	CALC / STUDY
Reference net helium lift	≈ 1.061 kg/m ³ at sea-level reference densities	CALC
Static buoyancy contribution target	~40% of design weight at reference condition	TARGET / STUDY
Residual rotor-supported weight	~60% of design weight ≈ 12,000 kg	CALC
Primary lift	Helium buoyancy + powered rotors for hover, control and forward flight	CONCEPT
Rotor architecture	Distributed or paired powered rotors; final count OPEN	STUDY
Occupancy	10–12 persons	TARGET / STUDY
Survival capsule mass target	5,200 kg	TARGET / STUDY
Landing supports	8 short, thick fixed mechanical supports integrated with capsule	TARGET
Range target	1,000 km	TARGET
Recovery	Capsule multi-canopy recovery; helium envelope emergency behavior separately analyzed	STUDY
Retractable landing gear	None	LOCKED CONCEPT

3. Preliminary Engineering Calculations

At sea-level reference densities, approximate net helium buoyancy is $1.225 - 0.164 = 1.061$ kg/m³ before envelope, ballonets, temperature and altitude corrections.

To support 40% of a 20,000 kg design weight statically, ideal gross helium volume is approximately $8,000/1.061 = 7,540$ m³.

The helium envelope itself has mass and structural drag; therefore the quoted volume is not a final aircraft sizing.

At altitude and temperature changes, both air density and helium density change; buoyancy therefore varies with operating condition.

Dynamic rotor lift remains essential for takeoff/landing control, maneuvering, wind response and any condition in which static buoyancy is insufficient.

Envelope puncture, fire, loss of helium, ballonets/pressure management, gust loads and ground handling become major safety drivers.

4. Multi-Layer Survival Architecture

L1. Automatic emergency detection and initiation.

L2. Independent backup initiation with segregated power/sensing.

L3. Independent manual crew authority.

L4. Deterministic Safety Kernel for rotor thrust, envelope pressure, attitude, wind/airspeed, structural and separation authorization.

- L5.** Helium-envelope health monitoring: pressure, temperature, volume/shape proxies and leakage/puncture indicators.
- L6.** Compartmentalized helium cells where feasible to reduce single-puncture loss.
- L7.** Fire/thermal isolation between propulsion, fuel/energy systems and helium envelope.
- L8.** Occupied survival capsule with independent power, recovery and eight-point impact attenuation.
- L9.** Selective separation of hazardous propulsion/fuel hardware only where trajectory and residual buoyancy are safe.
- L10.** Post-landing stabilization, locator, communications and rescue.

5. Variant-Specific Safety / Failure Cases

- Envelope rupture or progressive helium leakage.
- Loss of buoyancy and transition to higher rotor-supported weight.
- High wind/gust and mooring/ground-handling loads.
- Envelope/rotor collision or entanglement.
- Thermal effects and material degradation of the envelope.
- Fuel/engine fire near the helium structure.
- Loss of propulsion combined with low buoyancy margin.
- Capsule separation changing CG, buoyancy distribution and aerodynamic stability.
- Parachute deployment around the envelope must be studied for entanglement and inflation loads.
- Large-envelope ground impact, rollover and rescue-access hazards.
- Volume, drag, structural mass and manufacturing feasibility must be closed before claiming range or payload.

6. Core System Requirements

- 01.** Helium buoyancy shall be treated as supplementary lift, not as a sole survival mechanism.
- 02.** Envelope health monitoring shall detect defined pressure/leakage degradation states.
- 03.** The design shall define degraded buoyancy operating envelopes and minimum rotor-lift reserve.
- 04.** Envelope compartmentalization shall be studied for tolerance to localized damage.
- 05.** Propulsion and fuel systems shall be segregated from the helium envelope to reduce common-cause fire hazards.
- 06.** Capsule separation shall include updated CG, inertia and buoyancy state before recovery authorization.
- 07.** Safety Kernel shall inhibit actions that create unsafe envelope/rotor/parachute interaction.
- 08.** 1,000 km range remains conditional on aerodynamic drag, propulsion power, helium-system mass and mission fuel closure.

7. Verification and Development Gates

G0 architecture freeze → G1 mass/CG/inertia and analytical closure → G2 subsystem demonstrations → G3 integrated ground demonstration → G4 6-DOF/dynamic simulation correlation → G5 structural/crash testing → G6 recovery-system testing → G7 flight development → G8 certification/qualification evidence.

Variant-specific evidence must include performance closure, structural loads, propulsion/rotor or lift-system analysis, emergency-state dynamics, selective separation trajectories, capsule clearance, recovery deployment, impact attenuation and common-cause/independence analysis.

8. IP Position

This variant may strengthen the breadth and commercial applicability of the LSH platform if the variants share a genuine common survival architecture while solving materially different vehicle-level problems. It does not, by itself, establish novelty, patentability, freedom-to-operate or market value. Those require prior-art and legal analysis.

9. Critical Open Items

- Envelope rupture or progressive helium leakage. [OPEN]
- Loss of buoyancy and transition to higher rotor-supported weight. [OPEN]
- High wind/gust and mooring/ground-handling loads. [OPEN]
- Envelope/rotor collision or entanglement. [OPEN]
- Thermal effects and material degradation of the envelope. [OPEN]
- Fuel/engine fire near the helium structure. [OPEN]
- Loss of propulsion combined with low buoyancy margin. [OPEN]

- Capsule separation changing CG, buoyancy distribution and aerodynamic stability. [OPEN]
- Parachute deployment around the envelope must be studied for entanglement and inflation loads. [OPEN]
- Large-envelope ground impact, rollover and rescue-access hazards. [OPEN]
- Volume, drag, structural mass and manufacturing feasibility must be closed before claiming range or payload. [OPEN]

10. Engineering Integrity Note

No target in this document shall be presented as achieved. Detailed design requires aerodynamic/propulsive modelling, structural finite-element analysis, 6-DOF simulation, component testing, integrated ground testing and controlled flight testing as applicable.